

General Digital Currency Circulation Model

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Abstract

We describe the influence of digital currency on the currency circulation of sovereign countries, the governments' behaviors, and the world trade system from three aspects: individual transactions, national regulation, and world trade.

To describe clearly the behavior of individuals and countries from micro to macro perspectives, we divide the analytical framework into three submodels: Search and Matching Model, Long-Term Government Behavior Model, and Supranational Monetary System Model.

First, we use Fisher's equation and Keynes's theory of currency demand to analyze the pricing and risk characteristics of digital currency. The theory shows that the volatility of digital currency originates from speculative demand that is too maneuverable, while the transactional demand is insufficient because too few individuals trust digital currency. Therefore, expanding transactional demand and acceptance will help stabilize the currency value and enable it to act better as a trading medium.

Then we extend the Diamond-Mortensen-Pissarides Model (DMP) and established a Search and Matching Model for holding digital and fiat currencies in the market. We prove that there exists an equilibrium in the absence of external non-economic factors, that is, the proportion of people accepting digital to fiat currency will converge to a fixed range. But there is also a long-term situation in which some factors gradually expel the use of either fiat or digital currency.

We refer to the actual economic situation of more than 130 major countries in the world, substitute relevant parameter estimates, evaluate the possible currency-holding patterns of these countries without policy intervention, and conclude that the more-developed and open economies will accept

the coexistence of two currencies. The instability of currency value and other factors may cause a state to abandon fiat currency.

We set up a Long-Term Government Behavior Model to measure the government's regulatory behavior. The government regulates the proportion of currency use through the difference of taxation between the two payment modes, resulting in a cost difference for different currency users. On the one hand, the high liquidity of digital currency will promote the emergence of transactions, while at the same time it will increase the matching efficiency in transactions, which will motivate the government to promote the use of it to increase the total tax base. On the other hand, due to the additional regulatory costs for digital currency and the possible social losses caused by illegal transactions, the government also needs to regulate the development of digital currency to some extent.

Then we use the Profit Margin Model under Exchange Rate Fluctuation to conclude that under the hypothesis of relatively low volatility (that is, according to the theory of currency demand, the increase of demand for digital currency transactions will increase its stability), digital currency transactions will become the mainstream of international trade.

We establish a Supranational Monetary System Model, in which digital currency is controlled by a supranational group and distributed to countries according to foreign exchange demand. When the world circulates it publicly, digital currency will achieve lower price volatility. At that point, world trade will be settled mainly by digital currency. However, due to differences in development, fiat currencies will still be retained and independent monetary policies will be formulated among countries. Central banks of all countries have the tendency to control the balance of trade and capital flow; they will help countries formulate sound and positive economic development programs, promote the free flow of capital, and tap the most potential growth point of economic investment.

Finally, we discuss the impact of digital currency on the banking industry and the mechanism of long-term restructuring. The high liquidity and informatization of digital currency will lead to an oligopoly tendency of the banking industry and disappearance of the intermediary function of bank payments. Lending platforms similar to the banking industry will arise on the Internet. How to strengthen the supervision of illegal fund-raising and black-market transactions is a challenge for governments of all countries.

Introduction

Background

Ever since Satoshi Nakamoto released Bitcoin in 2009, digital currency trading has expanded rapidly. On the one hand, digital currencies have the advantages of strong liquidity, high secrecy, and low transaction costs. On the other hand, however, they are accompanied by a lack of national regulation. Also, central banks have found difficulties in macro-regulation and so on (though this may be part of the liberal ideal that Nakamoto wants to achieve).

Undeniably, almost all nations are stepping up establishment of a digital-currency trading mechanism, trying to bring digital currency into their regulatory system but also to maintain the dominance of their own sovereign *fiat currency*¹. With Internet transactions, digital currency has characteristics of a global currency, and its ability to promote global trade and capital circulation is better than that of regulated sovereign currencies. Whether the benefits of using digital currency in domestic trade and global trade can exceed the costs is a topic of concern.

Restatement of the Problem

We measure the impact of digital currencies on a country's fiat currency and on financial markets at the individual, national, and global levels.

We will study the motivation of nations to limit or encourage the development of digital currencies, regulatory means and objectives, as well as the monetary behavior game among countries.

Overview of Our Work

We build up models of transaction and regulation of fiat and digital currencies at three levels.

In the transactions between individuals, different forms of equilibrium will be achieved (e.g., natural exclusion of the fiat currency, exclusion of digital currency, or acceptance of two currencies at the same time), because of the difference in the degree of recognition and ease of circulation.

To measure the degree of recognition of fiat currency and digital currency, we draw lessons from the idea of Diamond-Mortensen-Pissarides Model of Search and Development (DMP) model [Morton and Pissarides 1994]. We introduce into the model the matching efficiency of trades, economic development, and other factors.

¹Fiat currency is government-issued currency whose value depends on the backing and stability of the government, since it does not have any backing in a physical commodity such as gold.

At the national level, we examine factors that contribute to national income (tax, regulation, and cost control) to establish a model to maximize national revenue, so as to evaluate the motivation of a country to encourage or restrict development of digital currency.

For monetary circulation in international trade, we build a Universal Digital Currency System Model based on Mundell's theory of Optimal Currency Areas (OCA) [Mundell 1961; 1968].

Under reasonable assumptions, we can prove that:

To reduce the risk of trading volatility, normal trade in goods will rely on digital currency in circulation worldwide rather than on the fiat currencies of countries.

At the same time, a supranational financial institution (similar to the current European Central Bank but more independent) will be created to regulate national trade and capital flows, but nations will still have the ability to use sovereign currencies within their borders and to carry out effective macroeconomic regulation.

This institution will make up for the weakness of monetary policy and fiscal policy under the fixed floating exchange rate system in the Mundell-Fleming-Dornbusch Model (MFD) with only sovereign currencies [Fleming 1962].

Finally, based on the impact of the decentralization of digital money on commercial banks, we explore the long-term trend of the evolution of commercial banks and policies.

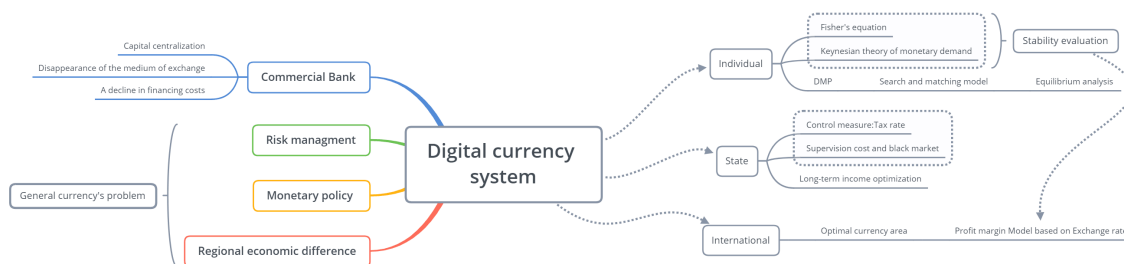


Figure 1. Overview of our work.

General Assumptions and Justifications

Assumptions

- The digital currency is widely distributed all over the world, with characteristics of decentralization, constant circulation, and anonymity.
- Everyone, including countries, is independent and free to choose currency without interference from others

- Digital currency and a country's fiat currency are relatively independent, and fluctuations in the value of one do not affect the other.
- Although digital currency is anonymous, countries can trace every transaction with advanced technique and at higher cost.

Table 1.
Parameter List.

Parameter	Description
$M_{\text{dig}} (M_{\text{fiat}})$	Monetary supply of digital currency (fiat currency)
ε_{dig}	Exchange rate of digital currency to fiat currency
P	Price of goods
Y	Quantity of goods
v	Currency circulation speed
$d (f)$	Proportion of digital currency (fiat currency) owners
$p_{\text{o-d}} (p_{\text{o-f}})$	Probability of matching a person without any currency with a person with fiat currency (digital currency)
$\mu_i, (\lambda_i)$	Probability in country i of accepting digital currency (fiat currency)
$V_{\text{d}} (V_{\text{f}})$	Value of digital currency (fiat currency)
δ	Measure of diversity of market demand
$pc_{\text{d}} (pc_{\text{f}})$	Cost of saving digital currency (fiat currency)
g_t	Revenue of government in year t
r	Real interest rate

Analysis and Model Building

What gave rise to the popularity of digital currency? What impedes its use?

Digital currency is based on blockchain technology, a distributed accounting method that does not require transactions to be confirmed by a third institution. Transfer is person to person; it is just a piece of information in the Internet ocean. Currency is no longer held by banks.

The convenience and concealment of transactions are the source of the demand for digital currency transactions. On the other hand, currency is a “medium of exchange, measure of value and value of storage” [Krugman 1984].

Digital currency is unlikely to be guaranteed by just a single country because of its global liquidity and lack of sovereign guarantee. Digital currency itself is not real goods of usable value, so it is not accepted by most people for most trading occasions. As a result, the trading demand for digital currency will be volatile, since its value exists only in the recognition by the populace. At the same time, high circulation of digital currency will help its speculative demand rise, making it unable to perform fully the functions of a currency as a medium of exchange and store for value.

Currency Model at the Individual Level

Digital Currencies: Price and Risk

We focus on a single domestic market, in which there is circulation of fiat currency of value M_{fiat} and also a digital currency in amount M_{dig} , with an exchange rate of digital currency against fiat currency of ε_{dig} . Thus, the value (supply) of digital currency in fiat currency

$$M_{\text{dig}}^s = \varepsilon_{\text{dig}} M_{\text{dig}}.$$

We divide the demand for digital currency into two parts: transactional and speculative:

$$M_{\text{dig}}^d = M_{\text{dig}}^{\text{business}} + M_{\text{dig}}^{\text{speculate}}.$$

According to Fisher's *quantity theory of money* [Fisher 1911], we have his *equation of exchange*

$$PY = Mv,$$

where

- P is the price level of goods in a currency,
- Y is the quantity of goods sold using the currency,
- M is the amount of the currency in circulation, and
- v is the velocity of currency circulation.

Due to the technical characteristics of digital currency, the velocity of its circulation will be a fixed value when the amount in circulation is controlled. As a result, the price of goods measured by digital currency will fall as transactions expand, so more goods will be traded in the currency. An increase in speculative demand for digital currency will trigger a rise in the price of goods measured in it. When the digital currency is not widely recognized, the quantity Y of goods is small, and the value volatility risk of the digital currency is expected to be greater.

Later we verify that when digital currency becomes the means for settlement of global trade, the universal recognition of digital currency leads to a more stable exchange rate than for a fiat currency, thus deepening the dependence of international trade on digital currency.

Combined with the theory of *speculative monetary demand* [Keynes 1965; Reddaway 1964] and the nature of digital currency itself, the speculative demand for digital currency is positively related to the exchange rate of the currency to fiat currency, the expected interest rate, and the increasing attention of the crowd. As the concept of digital currency expands, its value as an investment will increase, that is, an increase of speculative demand will push up its price.

Search and Matching Model under a Dual Monetary System

Model Specification

Suppose that in a closed economy within one single period, there are specific buyers and sellers, for which:

- The proportion of buyers holding fiat currencies is f and the proportion holding digital currency is d . No one holds both. The proportion of sellers (those who do not hold any currency) is $1 - d - f$.
- Each buyer is a representative agent (i.e., all buyers are identical), and each seller is also a representative agent.
- Everyone in the market carries out only one search and pairing when doing trades, and the searching process satisfies:
 - the larger the economic volume, the larger the market size;
 - the more efficient the pairing, the higher the matching efficiency; and
 - the more diversified the market demand, the more likely a match will succeed.

For simplicity, we set the matching efficiency to 1; so in the case of a random match between buyer and seller (see **Figure 2**), the probabilities of matching a seller with no currencies to a buyer holding fiat currency (p_{o-f}) or digital currency (p_{o-d}) are

$$p_{o-f} = \min \left\{ 1, \frac{f}{1 - f - d} \right\}, \quad p_{o-d} = \min \left\{ 1, \frac{d}{1 - f - d} \right\}.$$

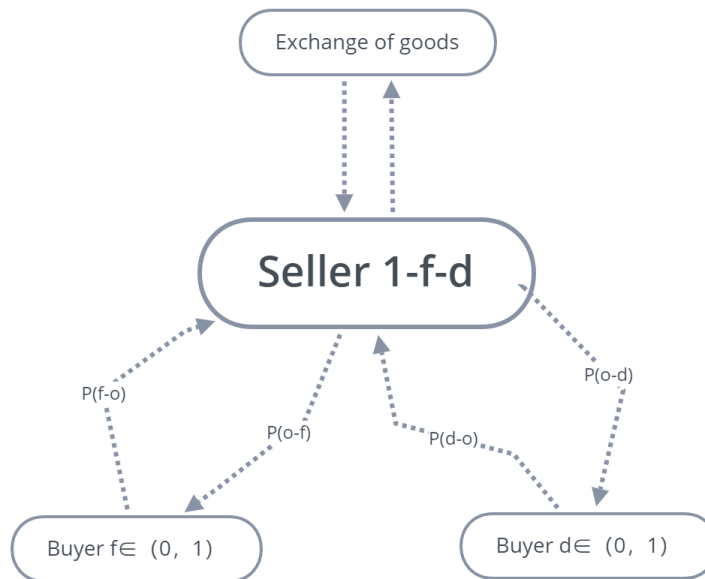


Figure 2. Matching relation.

The probabilities of matching a buyer holding fiat currency (p_{f-o}) or digital currency (p_{d-o}) to a seller not holding any currency are

$$p_{f-o} = \min \left\{ 1, \frac{1-f-d}{f} \right\}, \quad p_{d-o} = \min \left\{ 1, \frac{1-f-d}{d} \right\}.$$

We set the probability of acceptance of fiat currency to be λ and that of digital currency to be μ ; and V_o , V_f , and V_d are the values (in terms of fiat money) held by a person with no currency, fiat currency, and digital currency. Therefore, we have the following equations:

$$rV_o = (1 - p_{o-f} - p_{o-d})\delta^2(u - c) + p_{o-f}\lambda\delta(V_f - V_o - c) + p_{o-d}\mu\delta(V_d - V_o - c) \quad (1)$$

$$rV_f = p_{f-o}\lambda\delta(u + V_o - V_f) - pc_f, \quad (2)$$

$$rV_d = p_{d-o}\mu\delta(u + V_o - V_d) - pc_d, \quad (3)$$

where

- r is the interest rate,
- pc_f and pc_d are the costs of storing fiat currency and digital currency, and
- δ is a measure of diversity of market demand.

The equation indicates that conducting the transaction and refusing it are the same, for one can preserve the value of the goods and wait for the next transaction.

The first part of (1) is a barter that happens with a matching failure; the second part deals with the exchange between sellers with no currency and buyers holding fiat currency and digital currency respectively. Formulas (2)–(3) are from the angles of a fiat-currency holder and a digital-currency holder.

Equilibrium State of the Model

According to (1), when $V_f > V_o + c$, all sellers will accept fiat currency in order to maximize their profits ($\lambda = 1$). Contrarily, when $V_f < V_o + c$, all sellers will reject fiat currency ($\lambda = 0$). And when $V_f = V_o + c$, rejection and acceptance of fiat currency are the same ($0 < \lambda < 1$).

Similar relationships hold for digital currency, V_d vs. $V_o + c$, and the value of μ .

When the proportion of people receiving fiat currency is balanced, we have

$$\mu = 1 \quad \text{or} \quad \mu = \hat{\mu},$$

where solving (1) gives

$$\hat{\mu} = \begin{cases} A + \frac{(1 - p_{o-f} - p_{o-d})\delta}{p_{d-o}} + \frac{rc + pc_{d-o}}{p_{d-o}\delta(u - c)}, & \lambda < 1; \\ A + \frac{p_{o-f}(V_f - V_o - c)}{p_{d-o}(u - c)}, & \lambda = 1. \end{cases} \quad (4)$$

Similarly, when the proportion of people who accept digital currency in the market reaches equilibrium, corresponding equations hold for $\hat{\lambda}$ for the cases $\mu < 1$ and $\mu = 1$:

$$\hat{\lambda} = \begin{cases} B + \frac{(1 - p_{o-f} - p_{o-d})\delta}{p_{f-o}} + \frac{rc + pc_{f-o}}{p_{f-o}\delta(u - c)}, & \mu < 1; \\ B + \frac{p_{o-d}(V_d - V_o - c)}{p_{f-o}(u - c)}, & \mu = 1. \end{cases} \quad (5)$$

From the results of the above model, it can be seen that:

- When a currency (fiat or digital) is in market equilibrium, the other currency can also maintain a certain level of acceptance in the market.
- When one currency has reached equilibrium and the acceptance of the second one is lower than the theoretical equilibrium level, the low liquidity of the second currency will inevitably lead a wider population to abandon its use, and it will gradually be marginalized.
- When one currency has reached equilibrium and the acceptance of the second one is higher than the theoretical equilibrium level, the second one will fluctuate between a stable state of partially approved and fully approved until a balance is reached.

Due to the low storage cost of digital currency, there is no risk of management cost and hyperinflation, thus the possible equilibrium level of the fiat currency may be quite low.

Model for Maximizing Government Tax Revenue

We assume that the government's goal is to maximize its revenue over the long term; so in order to expand the tax base, the government is motivated to expand the economy. The expansion of economic activity benefits the improvement of searching efficiency.

National Cost-Benefit Analysis

Promotion of digital currency will increase its adoption and wider acceptance of it will contribute to economic development and thus expand the tax base.

At the same time, the rise in public acceptance of digital currency will reduce the use of fiat currency, forcing the government's tax base to shift from fiat currency transactions to digital currency transactions. But overall, as trade develops, the total tax base will increase, and tax revenues will rise.

In terms of transaction costs, digital currency has higher liquidity and concealment. The government's regulatory costs will rise with the volume of transactions using digital currency, and the widespread use of digital currency will deprive the country of its monetary policy controls over the macro economy. This potential systemic risk also needs to be factored into the total cost.

Government Intertemporal Tax Model

Because of the costs of monitoring tax evasion in digital currency and the risk of losing independent monetary policy, the government needs to impose different tax rates on the two payment modes to meet the needs of macro-control; on the other hand, the government has different regulatory costs for the two forms of payment, also leading to different tax rates.

GDP levels will rise as dependence on digital currency intensifies (although some of this growth may occur in illegal practices such as black-market transactions), and the increase in total trading volume will lead to greater probabilities of market matching. The tax base is expanded in terms of total income, so the total tax revenue will expand even if the tax revenue is transferred into digital currency.

Since willingness to change the form of payment is influenced by time, under the goal of maximization government revenue, the tax rates based on the shares of the current fiat currency and digital currency transactions can only be optimized in the long run. Our intertemporal model is that the government tries to maximize

$$\sum_{t=0}^n \frac{g_t}{(1+r)^t},$$

where g_t is tax revenue in year t , r is the rate of interest, and n is the time horizon. In our intertemporal model, the government continuously “induces” transactions to take an optimal proportion of payment means by selecting different tax rates in each period, until the total benefit of the discounted government tax is the greatest, after deducting the costs of supervision and risk.

International Digital Currency Model

We extend the use of digital currency to the global domain. We also prove that digital currency transactions will be used in large numbers in the payments in international trade, to reduce the risk of uncertainty arising from exchange rate fluctuations. We extend the model of Bacchetta and Wincoop [2002] about fluctuation of import and export exchange rates to the regulation mechanism of foreign exchange stability of digital currency as a third-party currency.

[EDITOR'S NOTE: We omit details of some subsequent arguments.]

Import and Export Profit under Exchange Rate Fluctuation

When two countries use one of their national fiat currencies for trade, optimization of profits means that the countries will choose the home currency of the exporter as the unit of valuation.

When the exchange rate fluctuation of the digital currency to the importing country is smaller than that between the exporting country and the importing country, the highest expected profit income will be obtained if the digital currency is chosen as the unit of valuation. Therefore, when the uncertain risk of digital currency can be effectively controlled, the choice of digital currency can promote effective trade between countries.

Supranational Monetary System Model at the World Level

Mundell [1961] and McKinnon [1963] put forward the theory of *optimal currency area* (OCA): that countries should abandon issuing sovereign currencies and establish supranational central banks in groups of countries with similar economic development patterns.

However, there are great differences in the economic situations of various countries, and giving a country's fiat currency does not take into account the importance of independent monetary policy to its domestic economy.

The supranational monetary system to be established will maintain the two-track parallel state of the global digital currency and the sovereign fiat currency. Moreover, the high liquidity of digital currency also gives the agency issuing the global common currency a more precise and flexible means of regulation and control. Using digital currency as the form of international circulation can still allow the power of independent choice of monetary policy in member countries.

Basic Hypothesis, Mechanism Analysis, and Model Construction

Assume that:

- A country will still issue and use its sovereign fiat currency f_i , with supply $M(f_i)$ and value $V(f_i)$.
- Digital currency is issued by an international financial institution, which distributes it to central banks in proportion to the needs of each country, and which is then issued by central banks in each country. The digital currency supply is expressed as $M(d_i)$.
- The digital currency will circulate globally, and its value will remain stable around the globe (i.e., its exchange rate will always be 1 between countries). The speed of circulation between countries will also remain unchanged, namely:

$$v(d) = v(d_{ij}), \quad i \neq j.$$

- The goal of the supranational financial institution is to keep the value of digital currency stable, with a low inflation rate, following Fisher's equation:

$$PY = Mv.$$

The institution estimates global output in order to determine the total supply of the global digital currency at a given velocity level:

$$\sum M(f_i) = \frac{PV}{v}.$$

From the results of our model, when the exchange rate volatility of the universal digital currency is less than that of each country (so that the total value of the global digital currency in circulation is nearly constant, hence conducive to maintaining the relative stability of exchange rates), every country involved has the motivation to adopt digital currency in foreign trade. Each domestic situation can be modeled with the earlier Search Matching Model.

The country's total demand for digital currency comes from two sources: the demand for imported goods and services, and the total demand for investment in all other countries. Also, it depends on the aggregate global demand and the speed of circulation.

To maintain a stable value of the digital currency, the supranational financial institution will supply the digital currency in proportion to the demand in each country.

Analysis on the Behavior Patterns of Each Subject

The supply of international digital currency must be stable so as to ensure that the exchange rate of the digital currency against sovereign fiat

currencies is less volatile, so the supranational financial institution should set inflation targets, generally 1.5%–2.5% [Liu et al. 2006]. Since inflation expectation is an important determinant of the inflation rate itself, when the real interest rate is known, setting a reasonable nominal interest rate will guide the market to adjust to the target inflation rate.

On the other hand, the supranational financial institution needs to evaluate and regulate the trade balance of each country. It also needs to control the surplus or deficit of the net capital inflow or outflow of each country.

Generally speaking, after generating a trade surplus, capital will flow out in order to realize the full use of capital; in this case, the supranational financial institution does not need to adjust the money supply to the country. If the country has a trade deficit and capital inflow, it means that the country has more pessimistic economic expectations, so the supranational financial institution will reduce the country's money supply and regulate the country's deficit in order to achieve the smooth operation of international trade.

Countries' central banks will issue their own sovereign fiat currencies and maintain a floating exchange rate relationship with general digital currencies.

Due to the use of universal digital currency in international trade, countries will focus on balance of payments when formulating foreign trade strategies. The share system of digital currency allocated by the supranational financial institution will make countries balance the relationship between capital flow and trade surplus (deficit).

If a country has a large trade deficit and there is no investment opportunity in the country, to attract foreign funds, the country may have to give up the power to issue fiat currency. Therefore, consciously adjusting the balance of payments will be an important goal of the central banks of all countries.

Due to the differing levels of inflation and real interest rates in countries, according to *purchasing power parity* (PPP) theory [Cassel 1916; Froot and Rogoff 1996], the exchange rate will change with the real interest rates and inflation rates at home and abroad. The assumptions of PPP can be more effectively realized by digital currency because of its high liquidity and high acceptance of global sharing.

Each country can manage the inflation and nominal interest rate of its sovereign fiat currency independently to realize independence of monetary policy. However, a change in the interest and inflation rates of the national currency will also lead to exchange rate fluctuation between the national currency and the general digital currency. This kind of exchange rate fluctuation can promote the export of domestic commodities in national currency and realize different trade strategies.

The difference between the real interest rate of each country and the others also leads to a difference of capital inflows of general digital currency capital to different countries.

In underdeveloped areas or domestic projects with better investment opportunities, due to the relatively high real interest rate (eliminating risk premium) and the wide application of a unified universally recognized digital currency, capital will rapidly enter a country and take up investment opportunities, which will help to distribute capital evenly among profitable projects and increase global investment efficiency.

Empirical Model

Empirical Study Based on Micro-model

Theoretical Restatement

Based on the theoretical model above, we can predict the future use of currency options in 130 countries.

We distinguish two extreme situations, Zimbabwe and the United States.

- Zimbabwe has given up issuance of a sovereign fiat currency, so its probability of accepting digital currency will most likely to be 100% when digital currencies emerge. So we assume that Zimbabwe is now in equilibrium with foreign currencies. Thus, $\mu = 1$, and the probability in Zimbabwe of accepting fiat currency satisfies (5) for $\hat{\lambda}$.

After calculating the critical value for Zimbabwe, we use it to predict the use of fiat currency in other countries. If the probability of accepting fiat currency in a country is lower than $\hat{\lambda}$ for Zimbabwe, it means that the country's fiat currency cannot reach an equilibrium. So if the country is using just fiat currency, that won't last long. Contrarily, if a country's acceptance of fiat currency is equal to or greater than $\hat{\lambda}$, it will end up in a fiat currency equilibrium.

- Since the dollar is the most widely accepted currency in the world today, we assume that the probability of acceptance of it by the American people is 100%; so the U.S. is now in fiat currency equilibrium, hence $\lambda = 1$. On the other hand, since the U.S. is the birthplace of a plentitude of digital currencies, its attitude towards them should be rather objective. So, we use the probability of the American people to accept digital currency as the threshold value for equilibrium with digital currencies. Thus, since $\lambda = 1$, the probability of adoption of digital currency in the U.S. satisfies (4) for $\hat{\mu}$.

Therefore, if the acceptance probability of digital currency in one country is lower than that of $\hat{\mu}$ for the U.S., the country cannot reach an equilibrium for digital currency, and the digital currency cannot circulate in that country for long. If the probability of acceptance of digital currency in a country is equal to or greater than $\hat{\mu}$ for the U.S., the country will end up with an equilibrium for digital currency, and the digital currency will be fully accepted after a period of time.

For the rest of the world, the probabilities of accepting fiat currency or digital currency are less than 100%. So for country i , the probabilities (λ_i) and μ_i of acceptance of fiat currency and of digital currency equal the expressions in (4) and (5) for the cases $\lambda < 1$ and $\mu < 1$ respectively.

By comparing these values with the thresholds for Zimbabwe and the U.S., we can find out which countries will adopt a new digital currency in global circulation and which countries will not.

Parameter Estimation and Results

To obtain the probabilities of acceptance of digital currency and fiat currency in each country, we need to estimate the relevant parameters using real data. To make it easier to calculate and without loss of accuracy, we assume that the benefit to each country through purchasing goods to be 1. Other parameter estimates are shown in **Table 2**.

Table 2.
Parameters and their representations for a country.

Parameter	Representation
r	Nominal interest rate
pc_{f-o}	fiat currency inflation rate
pc_{d-o}	Digital currency inflation rate
c	1 – national average profit margin
δ	Logarithm of GDP (gross domestic product)
p_{o-f}	fiat currency's share in the world monetary system
p_{o-d}	Bitcoin's share in the world monetary system

After using the parameter values for a country, we get the probabilities of acceptance of fiat currency and digital currency and compare those with the critical values.

Empirical Model Based on Econometric Methods

We construct an empirical model and do regression analysis of the factors that affect the demand for digital currency, as a verification and supplement to the results in the previous section.

Determinants of aggregate demand

As noted earlier, public demand M_{dig}^d for digital currency can be broken down into transactional demand $M_{dig}^{business}$ and speculative demand $M_{dig}^{speculate}$. What cannot be ignored is that risk factors will affect the efficiency of digital currency to assist the circulation of goods and the storage of value, and thus affect total monetary demand.

Aggregate demand for digital currency is a function of transactional demand (without considering risk), speculative demand (without considering risk), and risk itself.

Determinants of transactional demand

According to the Fisher equation $PY = Mv$, the transactional demand $M_{\text{dig}}^{\text{business}}$ is subject to the price level P , output level Y , and currency circulation speed v of the country.

We make the reasonable assumption that the speed v of circulation of digital currency in the world will be set, and its influence on the monetary demand of each country will be constant. As a result, the differences between countries in the demand for transactional currency are actually affected only by the level of output Y and the level of prices P . In the case of no risk, we should have

$$M_{\text{dig}}^{\text{business}} = M_1(P, Y).$$

Determinants of speculative demand

The current trading volume TR of digital currency will express well the public speculative demand for digital currency, and the degree IN of public interest in digital currency can reflect the potential speculative demand. Thus, for speculative demand for digital currency without risk, we have

$$M_{\text{dig}}^{\text{speculate}} = M_2(\text{TR}, \text{IN}).$$

Determinants of risk

We unravel the risk factors: legal risk, technical risk, and price risk.

- **Legal risk** will be determined by the attitude of a country's authorities to digital currencies, which we call (LE) (for "legislation"). It is an ordinal variable, with value: 0 (digital currency not allowed), 1 (development of digital currency restricted), 2 (active regulation), 3 (unclear attitude), or 4 (laissez-faire).
- **Technical risk** is influenced by transaction volume TR and the degree of network security SF (for "safety") of a country: With the increase of trading volume of digital currency, the possibility of technical failure will increase, and the difficulty of secrecy will also increase. Therefore, we expect the transaction volume to have a positive correlation with the risk. The higher the level of network security in a country, the less the possibility of technical risk, so the two have negative correlation.
- **Price risk** is expressed entirely by the uncertainty of the price of digital currency, and we assume that this variable is δPR (change in price).

So, risk is a function of LE, TR, SF, and PR.

Construction of the regression equation

Aggregate demand $M_{\text{dig}}^{\text{business}}$ in a country, i.e., the country's public acceptance of digital currency, is thus ultimately a function of price level P , output level Y , digital currency trading volume TR, public concern degree IN, national attitude LE, network security degree SF, and price fluctuation PR of digital currency:

$$\mu_i = \beta_0 + \beta_1 P_i + \beta_2 Y_i + \beta_3 \text{TR}_i + \beta_4 \text{IN}_i + \beta_5 \text{LE}_i + \beta_6 \text{SF}_i + \beta_7 \text{PR}_i + \epsilon_i,$$

where ϵ_i is a random error term.

Results of regression estimation

We use for each variable for country i :

- P_i , price level: the country's consumer price index.
- Y_i , output level: the country's GDP.
- TR, digital currency trading volume: Since Bitcoin is the best approximation so far of a universal digital currency), we use Bitcoin transactions in the country in the last 24 hours (in dollars; see CSPA Index [2019]).
- IN, degree of public concern: Bitcoin search heat of Google trends in the country in the past 24 hours.
- LE, the country's attitude toward digital currency: ordinal value 0 to 4.
- SF, degree of cybersecurity: the country's national ranking in the GCI (Global Network Security Index) to indicate the degree of cybersecurity.
- PR, price fluctuation of the digital currency: Bitcoin's percentage fluctuation against the national fiat currency in the last 24 hours.

Due to lack of data, we have full data from only 53 countries. We use Stata software to yield the ordinary-least-squares (OLS) estimates of the coefficients in the equation, with the results in **Table 3**.

The results are basically in line with expectations. The $R^2 = .28$ indicates that the dependent variables can explain the demand for digital currency to some extent. Trading demand P and speculative demand IN are significant at the levels 0.1 and 0.01 respectively. The three variables that are risk measurement factors are statistically insignificant (which may be related to insufficient sample size).

Final Results of the Empirical Model

We earlier calculated the probability of each country's acceptance of digital currency, and we used the regression model to verify and supplement the result.

Table 3.
Regression results.

Variable	Coefficient	Standard error
Constant	1.017***	(0.0655)
<i>P</i>	0.00697*	(0.00413)
<i>Y</i>	0.0106	(0.00640)
TR	0.0632	(0.0480)
IN	0.00469***	(0.000378)
LE	0.0113	(0.0101)
SF	−0.0295	(0.0315)
PR	−0.0185	(0.0285)

$N = 53, R^2 = .28,$

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

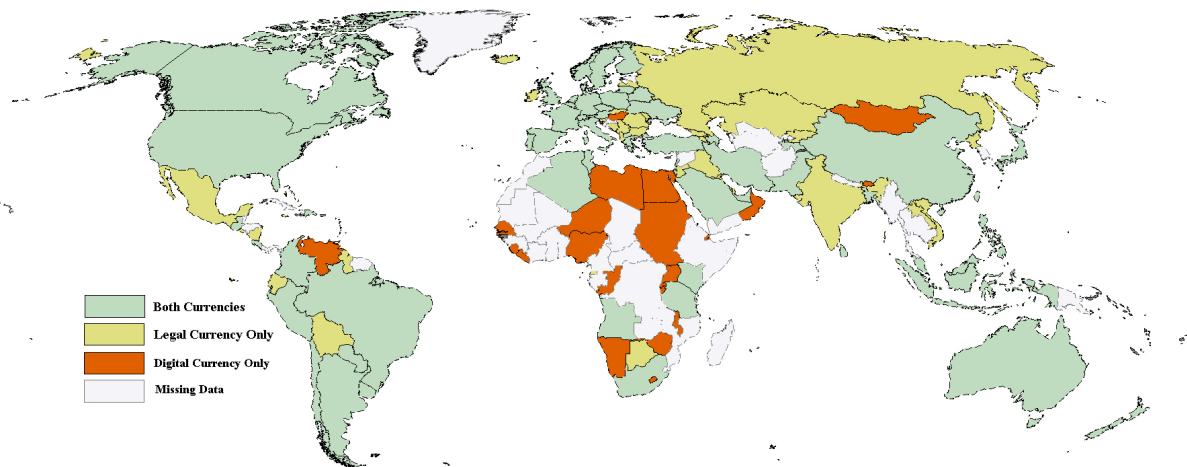


Figure 3. World monetary equilibrium.

Comparing with the critical values, we find the attitude of 130 countries toward the adoption of digital currency vs. the country's fiat currency. The result is shown in **Figure 3**.

When the new digital currency appears, major economies such as the U.S. and the European Union will adopt a relatively friendly attitude. These countries tend to occupy a dominant position in international trade. They already have an advantage in the performance of their fiat currencies, but the active adoption of digital currency can help them gain more trade benefits. Therefore, this part of the world can maintain its own fiat currencies while maintaining full acceptance of digital currencies.

Eastern countries, such as Russia and India, as well as some countries in the Americas, will adopt a relatively conservative attitude and will not accept digital currency. From a practical point of view, these countries have a high degree of control over their own currencies; and in order to avoid the impact of digital currencies on their own currencies, they will tend to

restrict the circulation of digital currency at home.

African countries, parts of Asia, and South America countries will fully adopt digital currency and abandon their own fiat currency. These governments have less control over their own currencies, and it will be better for their economic development to accept the widely-accepted universal digital currency instead of holding more volatile domestic currencies.

Empirical Study Based on Macro-model

Based on the macro model, we use Monte Carlo simulation to estimate the trading behavior of a country with a dual monetary policy. We also classify the international trade market into two categories:

- **The fiat currency market.** The operating mechanism of this currency market is the same as the current currency market. Commodity trade is carried out by converting currencies at exchange rates. The monetary policies of both countries directly affect the exchange rates. Only after the implementation of monetary policy or interest rate policy can two parties know the changes in their currencies' values and exchange rates. There is high volatility in the exchange rate.
- **The digital currency market.** Countries use digital currency to trade, so there is no currency exchange. The monetary policy and interest rate policy of the supranational financial institution will be announced ahead of time. Each country can adjust its own interest rate policy or monetary policy for fiat currency accordingly, which will benefit keeping the exchange rate relatively stable.

We fit the trade situation of the U.S. from 1994 to 2016 and get the corresponding fitting function to forecast the total trade volume of the U.S. for the next 10 years. If there is a digital currency acceptable to the whole world at that time, the U.S. will trade in both the fiat currency market and the digital currency market. We set the initial ratio randomly, and the annual trading ratio in the two currency markets is determined by the trading ratio in the previous year and the change in exchange rate.

After simulating with different initial parameters, we get the changes of the digital currency market share in the next 10 years. Through simulation, we find that the market share of digital currency will stabilize at about 65% regardless of the initial trading proportion.

But if the supranational central bank loses its regulatory capacity, the market share of digital currency will decline rapidly, to even lower than the initial share.

Further Discussion and Factors Outside the Model

Impact on Commercial Banks

We are concerned about the impact on the banking industry after the introduction of digital currency. The transaction form of digital currency greatly improves the efficiency of search and matching, the main business of commercial banks; they would gradually disappear from this service area. Though deposit and loan functions will continue to exist, there may be some trends:

- The scale effect of Internet finance will make commercial banks get rid of the coexistence of local banks such as community banks, city banks, and large banks at the international level.
- The promotion of informatization will reduce the cost of big banks and accelerate expansion of market share.
- Some “corners” that could not be covered will be occupied by convenient transactions of digital currency.
- Banks will rely more on innovative financial derivatives, so the regulatory problems of derivatives will become more prominent.
- A large number of financial services will still require banks to act as operators.
- Outside of large banks and financial institutions, the progress in digital currency transactions will reduce the costs of microlenders and informal lenders.

Exogenous Digital Currency Risk and Supervision

The reliability and privacy of digital currency deserve public attention. Present-day digital currency systems use blockchain technology for verification. However, losing an account password loses the digital currency in the account forever.

The decentralization of blockchain accounting makes manipulation and fraud difficult but also cannot fully adapt to the needs of complex reality. On the one hand, every transaction in the network is published. However, once the identity of an account holder is revealed, it will be difficult to prevent leakage of the holder’s transaction information. By contrast, the cost of national regulation must increase, which will correspondingly increase the cost of holding digital currency and reduce the acceptance of digital currency.

Independence of Monetary Policy

A two-currency monetary system will make it easier and faster to circumvent national monetary policy at the individual level. After the in-

roduction of the universal digital currency, if quantitative easing leads to instability in the value of a national currency, foreign capital will soon exit the country. In our Search and Matching Model, the use of digital currency increases rapidly. Eventually it will lead to a decline in the proportion of settlements accepted in fiat currency, resulting in a decline in the credibility of the government and the continued exclusion of the local fiat currency from the payment market.

State Control over Foreign Investment

The *infant industry theory* [List and Colwell 1856] holds that developing countries need to implement trade protectionism in the early stage of development of indigenous industries but participate in international competition when they are developed. Although this approach has been criticized by liberal economists, many countries follow such a policy. Countries are often afraid of foreign control over domestic arms industries, as well as infrastructure industries that are highly related to national interests. Therefore, the global liquidity of digital currency means that a country should abandon the protection of important industries in terms of foreign capital restriction and instead adopt direct administrative measures to intervene.

Gaps in the Economic Base

The establishment of the Eurozone and the euro currency is a great pioneering experiment in the theory of optimal currency, but practice in the Eurozone in the past 20 years shows that the economic differences among countries cannot be compensated by the rapid flow of capital suggested in the traditional model.

In our model, the generation of globalized digital currency is not fully applicable to the mechanism of capital flows to areas with higher nominal interest rates, because there may be a high-risk premium in less-developed areas that cannot be directly observed. In fact, capital entering underdeveloped countries in the euro area is mainly for speculative purposes, hence fails to fully promote development.

Strengths and Weaknesses

Strengths

- The core of our model is the acceptability of digital currency for individuals and manufacturers. Our models are flexible and compatible with blockchain technology.
- We could add more factors into our model to perfect it (for example, the exogenous factors that we mentioned).

- Based on the DMP model, we explain the relationship between the fiat currency of each country and a digital currency in global circulation. On this premise, we construct an influence mechanism that determines the country's attitude towards digital currency, which can explain some governments' monetary policies.
- Our models are based solidly in economic theory. Combining Fisher's equation with Keynes's theory of monetary demand, we make hypotheses about the flow mechanism of digital currency. Before the establishment of a supranational currency system, we demonstrate the prerequisites for the use of digital currency in international trade.

Weaknesses

- The framework of the model is relatively large and the relationships of submodels are relatively loose, which is not conducive to linkage analysis.
- Because a global decentralized digital financial market does not yet exist, our model has a relatively weak realistic foundation. At the same time, the theory of currency choice constructed at the macro level is relatively abstract.

Conclusion

We establish a model of trade-off between digital currency and fiat currency at the personal, domestic, and international levels. In the domestic part, we determine a dynamic equilibrium between a country's fiat currency and the digital currency, by simplifying market transactions. We determine an interval in which the proportions of the two currencies tend to stabilize.

We do an empirical analysis of the model. We set up the model parameters according to relevant economic theory and estimate the parameters by using OLS regression. The model can help us judge the likelihood of use of digital currency in major countries around the world when a new digital currency emerges. If a country has a dominant position in international trade and a stable domestic economy, it is more likely to adopt a monetary policy that uses both fiat and digital currencies. However, a country with a chaotic domestic economy and an unstable currency will tend to abandon its original fiat currency. On the other hand, if a country has a firm attitude towards domestic monetary control, and a lower degree of marketization or a more conservative attitude, it will tend to recognize fiat currency only.

In the international arena, we verify that with widespread recognition of digital currency and the existence of strict inflation controls, the risk of

digital currency is manageable. A smooth digital currency will play a major role in international trade. Central banks of countries will still have the power to issue sovereign fiat currency. They also can adjust the proportion of payment between the two currencies and formulate appropriate monetary policy; for example, a central bank can use quantitative easing to stimulate export. At the same time, the supranational monetary system will also force the country to limit its own liabilities and balance imports and exports. The model also demonstrates that digital monetary capital will flow rapidly into projects with higher real interest rates, promoting growth in the entire economy.

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POLICY RECOMMENDATION

To: National Leaders
 From: Team 1905127
 Subject: Policy recommendations on building a digital financial market
 Date: Jan 28, 2019

Dear National Leaders:

We offer you a perspective on the development of a global digital monetary system. Our team has collected the characteristics of digital currencies and blockchain technology, and combined it with the existing economic and monetary finance theories. We establish a mechanism analysis model from the personal, national, and international levels, trying to explain the trend of development of a digital currency system and the existing risk factors.

According to the circulation law of money, when not many transactions are covered by a digital currency, its speculative demand is far higher than its transactional demand. According to the Fisher equation, a lack of trades in a digital currency leads to high volatility. If a large proportion of people participate in transactions with a digital currency, the fluctuation in demand will become relatively more stable.

An individual may reject the use of one currency and choose another because of a difference in the degree of currency recognition. Ultimately, a currency's recognition is determined by whether sellers and buyers can benefit from using it. With an increase in recognition, the use of digital currency will be more frequent and equilibrium in its use will finally be

achieved. Therefore, either one form of currency will dominate, or else there will be long-term coexistence of two currencies. However, due to the lower transaction and storage costs of digital currencies, traders' acceptance of digital currencies will increase.

At the national level, government's goal is to maximize long-term tax income. The government can change the amounts of transactions settled in the two currencies by adjusting the tax rates in them and by guiding public recognition of the currency. Of course, if the transactional cost of digital currency is low, adoption of a lower tax rate for it will lead to an increase of the total national economy (not necessarily just normal economic transactions, but also smuggling, etc.). The matching efficiency of transactions will improve, which will increase taxes received. There will also be an increase in law enforcement costs, so the government needs to weigh the pros and cons.

At the international level, with wide use in trade settlement, a digital currency's stability will rise, so enterprises will suffer less fluctuation losses in exchange rates. Global trade settlement will thus be dominated by digital currency. With the establishment of a worldwide digital monetary control system, our central bank will still be able to conduct monetary policies independently, and will be more motivated to balance import, export and capital flows. Because of the free flow of digital money, capital will flow to more profitable places, which will benefit many previously neglected underdeveloped areas.

Therefore, in summary, we should give full play to the advantages of digital currency, while controlling the uncertainty of exchange rate and the risk of illegal transactions.

Sincerely,

Team 1905127

About the Authors



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